# Bayesian Geometric Inference: Generalizing the Riemannian Sector of General Relativity's Mathematics

### to Arbitrary Physics-Constrained State Estimation

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**Abstract**

We generalize the Riemannian sector of General Relativity's mathematical framework—specifically Synge's world function formulation—to arbitrary physics-constrained state estimation problems. We show that any system with physics constraints induces a Riemannian manifold where optimal estimation corresponds to geodesic projection, and we derive a world function (the *Modak-Walawalkar Distance*) that satisfies the same fundamental properties as Synge's formulation in the Riemannian (positive-definite metric) case.

Our framework extends Riemannian geometric inference to arbitrary $N$-dimensional state manifolds where metrics are learned through Bayesian inference rather than derived from field equations, enabling: (1) tunable dimensionality adapted to problem complexity ($N$-D latent spaces vs spacetime's fixed 4-D), (2) computational tractability via automatic differentiation replacing manual tensor calculus, (3) universal applicability to any physics constraints rather than gravity alone, and (4) real-time deployment in safety-critical systems.

We employ Bayesian Variational Autoencoders to learn manifold geometry, complementing Einstein's analytical approach with data-driven inference. We validate across two entirely distinct domains: electrochemical battery degradation (16 physics priors including Arrhenius kinetics, SEI growth, mechanical stress) and operational technology cybersecurity (15 security priors including vulnerability accumulation, exposure dynamics, attack propagation). Both exhibit identical geometric structures—including $\sqrt{t}$ growth laws, exponential acceleration factors, and stress accumulation—despite operating via completely different mechanisms.

This convergence provides empirical evidence that Riemannian geometric inference is universal across constrained systems. On battery state-of-health estimation, we achieve 20-200× speedup versus online PDE solving while providing multi-dimensional health scores with formal uncertainty guarantees. On OT cybersecurity, we provide attack likelihood predictions with component-wise vulnerability decomposition.

**Scope:** This work generalizes GR's Riemannian mathematics (spatial geometry, positive-definite metrics). GR's Lorentzian structure (causality, time ordering) is outside the present scope.

**Keywords:** Bayesian inference, Riemannian geometry, General Relativity, Synge world function, variational autoencoders, physics-informed machine learning, state estimation

# 1 Introduction

## 1.1 The Geometric Revolution in Physics

Einstein's General Relativity (1915) transformed our understanding of gravity through a profound insight: gravitational physics is geometry. Rather than forces acting in flat spacetime, mass-energy curves the fabric of 4-dimensional spacetime itself, and particles follow geodesics—shortest paths—through this curved manifold. The mathematical machinery developed to formalize this insight—metric tensors, Christoffel symbols, Riemann curvature, and Synge's world function—has remained largely confined to gravitational physics for over a century.

We ask: Is this geometric approach specific to gravity, or does it reflect a deeper principle?

## 1.2 Our Central Thesis

We generalize the Riemannian sector of General Relativity's mathematical framework to arbitrary physics-constrained state estimation. Specifically: Any dynamical system with physics constraints defines a Riemannian manifold, and optimal state estimation under these constraints corresponds to geodesic projection onto that manifold—exactly the mathematical structure used in GR's spatial geometry.

General Relativity's Riemannian sector consists of:

- Metric tensors $g_{ij}$ on spatial hypersurfaces
- Geodesic distance computations
- Synge's world function (in the Riemannian case)
- Christoffel symbols and curvature on positive-definite manifolds

We demonstrate that this Riemannian geometric machinery generalizes beyond spacetime to *any* physics-constrained inference problem, while acknowledging that GR's full Lorentzian structure (indefinite signature, causality, time ordering) is a separate mathematical layer requiring distinct treatment.

More precisely, we show that:

1. **Constraints induce geometry**: Physics constraints (conservation laws, constitutive relations, degradation dynamics) naturally induce Riemannian manifold structure in state space
2. **Bayesian inference learns metrics**: The metric tensor can be learned via Bayesian variational inference, complementing analytical derivation from field equations
3. **Arbitrary dimensionality**: The manifold dimension $N$ is tunable based on problem complexity, adapted to the specific system rather than fixed by fundamental physics
4. **Universal world function**: A world function measuring geodesic distance can be defined on any learned physics manifold, satisfying the same properties as Synge's formulation
5. **Computational tractability**: Automatic differentiation replaces manual tensor calculus, making the approach practical for real-world deployment

### 1.3 The Modak-Walawalkar Framework

We present a complete mathematical framework—the *Modak-Walawalkar Theory*—comprising:

- **Physics-Informed Variational Autoencoders (PI-VAE)**: Learn low-dimensional manifolds where all physics constraints are satisfied jointly
- **Learned Riemannian Metrics**: Metric tensors $g_{ij}(z)$ induced by physics priors through decoder Jacobian and importance weights
- **Modak-Walawalkar Distance**: A bi-scalar world function $\Omega_M(x, x')$ measuring deviation from physics manifold
- **Geometric Diagnostics**: Curvature tensors (Riemann, Ricci, scalar) computed via automatic differentiation
- **Uncertainty Quantification**: Van Vleck-type determinants providing formal confidence bounds

### 1.4 Comparison to General Relativity

Table 1 summarizes the mathematical correspondence and key distinctions.

Table 1: General Relativity vs Modak-Walawalkar Framework

| **Feature** | **GR** | **M-W** |
|---|---|---|
| Dimension | 4 (fixed) | $N$ (tunable) |
| Metric source | Field equations | Learned |
| Computation | Solve PDEs | Autodiff |
| Physics scope | Gravity | Universal |
| Real-time | No | Yes |
| **Signature** | **Lorentzian** $(-,+,+,+)$ | **Riemannian** $(+,+,+,\ldots)$ |
| **Dynamics** | **Energy-momentum** | **Learned posterior** |
| **Symmetry** | **Diffeomorphism inv.** | **Prior-dependent** |

*Note:* We generalize GR's Riemannian sector (positive-definite metrics, spatial geometry). Lorentzian structure is outside the present scope.

### 1.5 Contributions

Our specific contributions are:

1. **Theoretical**: Proof that learned Riemannian manifolds from Bayesian inference satisfy Synge-type world function properties (Theorems 1-4)
2. **Algorithmic**: Computational framework using automatic differentiation for all geometric quantities, eliminating manual tensor calculus
3. **Empirical**: Validation across two entirely distinct domains (electrochemistry and cybersecurity), demonstrating universality
4. **Practical**: Real-world deployable system achieving 20-200× speedup with formal uncertainty guarantees

### 1.6 Organization

Section 2 reviews related work in General Relativity, physics-informed ML, and geometric deep learning. Section 3 presents the mathematical framework. Section 4 details the Modak-Walawalkar Distance and its properties. Section 5 validates on battery degradation. Section 6 validates on OT cybersecurity. Section 7 discusses the geometric universality and future directions.

## 2 Background and Related Work

### 2.1 Synge's World Function in General Relativity

In General Relativity, Synge's world function [1] is a bi-scalar $\Omega(P, Q)$ measuring half the squared geodesic distance between spacetime points $P$ and $Q$:

$$\Omega_{\text{Synge}}(P, Q) = \frac{1}{2}(\lambda_1 - \lambda_0) \int_{\lambda_0}^{\lambda_1} g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \, d\lambda \tag{1}$$

This function satisfies fundamental properties [2, 3]:

- **Coincidence limit**: $\lim_{Q \to P} \Omega(P, Q) = 0$
- **Geodesic equation**: $\nabla_Q \Omega|_{Q=P}$ yields the geodesic tangent
- **Parameterization invariance**: Independent of curve parameterization

The Van Vleck determinant $\Delta(P, Q) = \det(-\nabla_P \nabla_Q \Omega)$ measures geodesic focusing and appears in semiclassical propagators [4].

**Limitation**: These constructions require known metric tensors derived from field equations—typically solvable only for highly symmetric spacetimes.

### 2.2 Physics-Informed Neural Networks

Physics-Informed Neural Networks (PINNs) [5] incorporate governing PDEs as soft constraints during training. However, PINNs require solving optimization problems at inference time, precluding real-time deployment. Our VAE approach learns the physics manifold once, then performs fast feedforward inference.

### 2.3 Riemannian Variational Autoencoders

Recent work [6, 7] applies Riemannian geometry to VAE latent spaces, with learned metrics capturing data geometry. Our contribution extends this by:

1. Deriving metrics from physics priors rather than data alone
2. Establishing formal connections to Synge's world function
3. Proving optimality for constrained state estimation
4. Demonstrating universality across domains

### 2.4 Kalman Filtering and State Estimation

Extended Kalman Filters (EKF) [8] combine physics models with measurements but assume Gaussian noise and require linearization. They lack:

- Representational capacity for high-dimensional nonlinear manifolds
- Unified treatment of multiple physics constraints
- Formal geometric structure with world functions

Sequential filter chains [9] apply physics corrections sequentially (Arrhenius → C-rate → aging), suffering from order dependence and inconsistent state predictions. Our manifold approach enforces joint constraint satisfaction.

## 3 Mathematical Framework

### 3.1 Problem Formulation

Consider a safety-critical system with state vector $\mathbf{x} \in \mathbb{R}^d$ comprising:

- $\mathbf{x}_{\text{obs}} \in \mathbb{R}^n$: Observable telemetry (sensor measurements)
- $\mathbf{x}_{\text{hidden}} \in \mathbb{R}^m$: Hidden states (unmeasured variables)

where $d = n + m$.

The system evolves according to physics:

$$\mathbf{x}_{t+1} = f_{\text{physics}}(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\theta}) + \mathbf{w}_t \tag{2}$$

where $\mathbf{u}_t$ are control inputs, $\boldsymbol{\theta}$ are system parameters, and $\mathbf{w}_t \sim \mathcal{N}(0, \mathbf{Q})$ is process noise.

**Goal**: Given partial noisy telemetry $\mathbf{y}_t$, estimate the full state $\mathbf{x}_t$ such that:

1. Estimates are physics-consistent
2. Match telemetry within sensor accuracy
3. Provide uncertainty bounds for safety margins

### 3.2 The Physics Manifold Hypothesis

**Definition 1** (Physics Manifold)**.** Physically valid states form a low-dimensional manifold $\mathcal{M} \subset \mathbb{R}^d$ defined by:

$$\mathcal{M} = \{\mathbf{x} \in \mathbb{R}^d : C_i(\mathbf{x}) = 0,\ i = 1, \ldots, K\} \tag{3}$$

where $\{C_i\}$ are physics constraint functionals (conservation laws, constitutive relations, etc.).

**Key insight**: Rather than applying constraints sequentially as filters, we learn the manifold $\mathcal{M}$ where all constraints are satisfied jointly.

### 3.3 Physics-Informed Variational Autoencoder

We employ a Variational Autoencoder [10] augmented with physics priors:

**Encoder**: $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}), \boldsymbol{\Sigma}_\phi(\mathbf{x}))$

**Decoder**: $p_\theta(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{z}), \boldsymbol{\Sigma}_\theta(\mathbf{z}))$

The decoder defines a mapping $D : \mathcal{Z} \to \mathbb{R}^d$ such that:

$$\mathcal{M} = \{D(\mathbf{z}) : \mathbf{z} \in \mathcal{Z}\} \tag{4}$$

Training objective:

$$\mathcal{L} = \mathbb{E}_q[\log p(\mathbf{x}|\mathbf{z})] - \beta \cdot D_{\text{KL}}[q(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})] + \lambda \mathcal{L}_{\text{physics}} \tag{5}$$

where $\mathcal{L}_{\text{physics}}$ encodes domain-specific constraints.

### 3.4 Bayesian Physics Priors

We implement physics constraints through Bayesian priors on parameters:

**Example (Batteries)**: Arrhenius temperature dependence

$$k(T) = A \exp\left(-\frac{E_a}{k_B T}\right) \tag{6}$$

becomes a prior on activation energy:

$$\log E_a \sim \mathcal{N}(\mu_{E_a}, \sigma_{E_a}^2) \tag{7}$$

**Example (Cybersecurity)**: Vulnerability accumulation

$$V(t) = k_{\text{vuln}}\sqrt{t} + \text{CVE}_{\text{critical}} \tag{8}$$

with prior:

$$k_{\text{vuln}} \sim \text{LogNormal}(\mu_k, \sigma_k^2) \tag{9}$$

This approach allows us to encode 15+ physics priors per domain without manual derivation of field equations.

### 3.5 Learned Riemannian Metric

**Definition 2** (Physics-Induced Metric)**.** The decoder $D : \mathcal{Z} \to \mathcal{M}$ induces a pullback metric on the latent space:

$$g_{ij}(\mathbf{z}) = \sum_{\alpha=1}^{d} \frac{\partial D^\alpha}{\partial z^i} \frac{\partial D^\alpha}{\partial z^j} \Phi_\alpha \tag{10}$$

where $\Phi_\alpha$ are physics-derived importance weights reflecting the physical significance of each state dimension.

Weights encode domain knowledge:

- Conservation laws (energy, mass, charge)
- Safety-critical variables (temperature limits, voltage bounds)
- Degradation indicators (SOH, resistance, CVE counts)

**Relationship to General Relativity**: This generalizes GR's Riemannian geometry:

1. **What We Generalize**: Positive-definite metrics $g_{ij}$, geodesics, Synge world function, Christoffel symbols, and curvature tensors. This mathematical machinery applies equally to spatial GR slices or arbitrary state manifolds.

2. **Key Differences**:
   - **Signature**: Riemannian $(+, +, +, \ldots)$ vs Lorentzian $(-, +, +, +)$
   - **Dynamics**: Learned from Bayesian priors vs derived from Einstein equations
   - **Source**: Data-driven vs field equations

3. **Universality**: The Riemannian formalism works for any constrained manifold—GR spatial slices, battery states, cybersecurity surfaces, etc.

### 3.6 Telemetry Preservation

To respect sensor accuracy, we introduce observation weights $\mathbf{w} \in [0,1]^d$:

- $w_i = 1.0$ for high-accuracy telemetry (preserve exactly)
- $w_i \ll 1$ for uncertain parameters (allow physics correction)

This forces the VAE to project uncertain measurements onto the physics manifold while respecting direct observations—a critical constraint for safety-critical deployment.

## 4 The Modak-Walawalkar Distance

### 4.1 Motivation: Two Formulations

We present two formulations suited to different manifold geometries:

**Definition 3** (Modak-Walawalkar-Euclidean Distance (Type I))**.** For smooth manifolds with low curvature:

$$\Omega_M^{(E)}(\mathbf{x}, \mathcal{M}) = \frac{1}{2}\|\mathbf{W}^{1/2} \cdot (\mathbf{x} - \Pi_{\mathcal{M}}(\mathbf{x}))\|_p^p \tag{11}$$

where $\mathbf{W} = \mathrm{diag}(\Phi_1, \dots, \Phi_d)$ and $\Pi_{\mathcal{M}}(\mathbf{x}) = D(E(\mathbf{x}))$ is the projection onto $\mathcal{M}$.

**Definition 4** (Modak-Walawalkar-Geodesic Distance (Type II))**.** For manifolds with sharp safety boundaries:

$$\Omega_M^{(G)}(\mathbf{x}, \mathbf{x}') = \frac{1}{2}\int_0^1 g_{ij}(\boldsymbol{\gamma}(\lambda))\frac{d\gamma^i}{d\lambda}\frac{d\gamma^j}{d\lambda}\, d\lambda \tag{12}$$

where $\boldsymbol{\gamma} : [0,1] \to \mathcal{Z}$ is the geodesic connecting $E(\mathbf{x})$ to $E(\mathbf{x}')$.

**Structural parallel to Synge**: The Type II formulation has identical mathematical structure to Synge's world function (Equation 1), despite being derived from entirely different physical reasoning. This convergence is not coincidental—it reflects a universal geometric structure underlying physics-constrained inference.

**Key distinction**: In GR, the metric $g_{\mu\nu}$ and its geodesics are *outputs* of Einstein's field equations. In our framework, the metric $g_{ij}$ and geodesics are *learned* from data subject to physics priors. Both yield world functions with the same formal properties (Theorems 1-3), demonstrating that this geometric structure transcends its gravitational origins.

### 4.2 Mathematical Properties

We now prove that the Modak-Walawalkar Distance satisfies the same fundamental properties as Synge's world function.

**Theorem 1** (Coincidence Limit)**.** *The Modak-Walawalkar-Geodesic Distance satisfies:*

$$\lim_{\mathbf{x}' \to \mathbf{x}} \Omega_M(\mathbf{x}, \mathbf{x}') = 0 \tag{13}$$

*with gradient condition:*

$$[\nabla_{\mathbf{x}'}\Omega_M]_{\mathbf{x}'=\mathbf{x}} = \mathbf{0} \tag{14}$$

*Proof.* As $\mathbf{x}' \to \mathbf{x}$, we have $\mathbf{z}' = E(\mathbf{x}') \to E(\mathbf{x}) = \mathbf{z}$ by continuity of the encoder. The geodesic $\boldsymbol{\gamma}$ connecting $\mathbf{z}$ to $\mathbf{z}'$ shrinks to a point, and the integral in Definition 4 vanishes. The gradient condition follows from the stationarity of geodesics at the endpoint. □

**Theorem 2** (Parameterization Invariance)**.** $\Omega_M(\mathbf{x}, \mathbf{x}')$ *is independent of the parameterization of the geodesic* $\boldsymbol{\gamma}$.

*Proof.* Let $\tilde{\lambda} = f(\lambda)$ be a reparameterization with $f(0) = 0$, $f(1) = 1$. The geodesic tangent transforms as $d\boldsymbol{\gamma}/d\tilde{\lambda} = (d\boldsymbol{\gamma}/d\lambda) \cdot (d\lambda/d\tilde{\lambda})$. Substituting:

$$\tilde{\Omega}_M = \frac{1}{2}\int_0^1 g_{ij}\frac{d\gamma^i}{d\tilde{\lambda}}\frac{d\gamma^j}{d\tilde{\lambda}}\,d\tilde{\lambda} \tag{15}$$

$$= \frac{1}{2}\int_0^1 g_{ij}\frac{d\gamma^i}{d\lambda}\frac{d\gamma^j}{d\lambda}\left(\frac{d\lambda}{d\tilde{\lambda}}\right)^2\frac{d\tilde{\lambda}}{d\lambda}\,d\lambda \tag{16}$$

$$= \frac{1}{2}\int_0^1 g_{ij}\frac{d\gamma^i}{d\lambda}\frac{d\gamma^j}{d\lambda}\,d\lambda = \Omega_M \tag{17}$$

where we used $d\lambda/d\tilde{\lambda} = (d\tilde{\lambda}/d\lambda)^{-1}$. □

**Theorem 3** (Geodesic Equation Correspondence)**.** *The gradient of $\Omega_M$ with respect to $\mathbf{x}'$ satisfies:*

$$\nabla_{\mathbf{x}'}\Omega_M = g_{ij}(\mathbf{z}')\dot{\gamma}^j \cdot \frac{\partial E}{\partial \mathbf{x}'} \tag{18}$$

*where $\dot{\boldsymbol{\gamma}}$ is the geodesic tangent at $\mathbf{z}'$.*

*Proof.* Following Synge [1], Section 2.3, the variation of the world function with respect to the endpoint yields the geodesic tangent vector, transported through the encoder Jacobian to the ambient space $\mathbb{R}^d$. □

### 4.3 The Universality Theorem

We now state our central theoretical contribution precisely:

**Theorem 4** (Riemannian Geometric Inference Universality)**.** *Let $\mathcal{M}$ be a smooth manifold embedded in $\mathbb{R}^d$ defined by physics constraints $\{C_i(\mathbf{x}) = 0\}$. Let $g_{ij}$ be a Riemannian (positive-definite) metric on $\mathcal{M}$ (either derived analytically or learned via Bayesian inference). Then there exists a bi-scalar world function $\Omega(\mathbf{x}, \mathbf{x}')$ satisfying:*

1. *Coincidence limit: $\Omega(\mathbf{x}, \mathbf{x}') \to 0$ as $\mathbf{x}' \to \mathbf{x}$*
2. *Geodesic equation: $\nabla\Omega$ yields geodesic tangents*
3. *Parameterization invariance*
4. *Van Vleck determinant structure*

*Furthermore, these properties hold independently of:*

- *The manifold dimension $N$*
- *How the metric was obtained (field equations vs learning)*
- *The specific physical domain (gravity, electrochemistry, cybersecurity)*

*This establishes that the Riemannian sector of GR's mathematics generalizes to arbitrary constrained inference.*

*Proof.* Follows from Theorems 1-3 and the general construction of geodesic distance on Riemannian manifolds. The key insight is that these properties depend only on the *existence* of a Riemannian metric, not on *how* that metric was derived or what physical system it describes. □

**Corollary 1** (General Relativity's Riemannian Sector as Special Case)**.** *General Relativity's spatial geometry corresponds to the special case where:*

1. *$\mathcal{M}$ is a 3-dimensional spatial hypersurface in spacetime*
2. *$g_{ij}$ is the induced Riemannian metric on that hypersurface*
3. *The metric satisfies constraints from Einstein's field equations*
4. *The world function is Synge's construction restricted to the Riemannian sector*

*The properties (1)-(4) in Theorem 4 hold for this case and for ANY other Riemannian manifold, demonstrating that this geometry generalizes beyond spacetime.*

**Corollary 2** (Empirical Convergence)**.** *The appearance of identical geometric structures ($\sqrt{t}$ growth laws, exponential acceleration, geodesic focusing) across unrelated physical domains (electrochemistry, cybersecurity, spatial GR) is not coincidental—it reflects the universal Riemannian structure identified in Theorem 4.*

### 4.4 Van Vleck Determinant Analogy

**Definition 5** (Modak-Walawalkar Determinant)**.**

$$\Delta_M(\mathbf{x}, \mathbf{x}') = \det(\mathbf{J}_E^T \cdot \mathbf{H}_\Omega \cdot \mathbf{J}_E) \tag{19}$$

where $\mathbf{J}_E$ is the encoder Jacobian and $\mathbf{H}_\Omega$ is the Hessian of $\Omega_M$ in latent space.

**Proposition 1** (Uncertainty Interpretation)**.** *$\Delta_M^{-1/2}$ provides a natural measure of prediction uncertainty, analogous to the role of Van Vleck determinant in semiclassical path integrals.*

This gives us formal uncertainty bounds:

$$\sigma_{\text{pred}} \propto \frac{1}{\sqrt{\Delta_M(\mathbf{x}, \Pi_{\mathcal{M}}(\mathbf{x}))}} \tag{20}$$

### 4.5 Component-Wise Decomposition

The Modak-Walawalkar Distance admits a natural decomposition:

$$\Omega_M = \sum_{\alpha=1}^{d} \Phi_\alpha \cdot \Omega_M^{(\alpha)} \tag{21}$$

This enables:

1. **Root cause analysis**: Identify which physical subsystem is anomalous
2. **Interpretable diagnostics**: Map anomaly scores to physical phenomena
3. **Targeted interventions**: Prioritize maintenance based on component contributions

## 5 Validation: Battery State Estimation

### 5.1 Domain Physics

Lithium-ion battery degradation involves complex coupled phenomena:

1. **Arrhenius Temperature Dependence**:

$$k(T) = A \exp\left(-\frac{E_a}{k_B T}\right) \tag{22}$$

2. **SEI Growth (Parabolic Law)**:

$$\delta_{\mathrm{SEI}}(t) = k_{\mathrm{SEI}}\sqrt{t} \cdot f(T, \mathrm{SOC}) \tag{23}$$

3. **Mechanical Stress**:

$$\sigma(t) = \sigma_0 + k_{\mathrm{stress}} \cdot \mathrm{cycles} \tag{24}$$

4. **Calendar Aging**:

$$\Delta Q_{\mathrm{cal}} = k_{\mathrm{cal}}\sqrt{t_{\mathrm{days}}} \tag{25}$$

5. **Lithium Plating Risk**:

$$P_{\mathrm{plating}} = f(T, C\text{-rate}, \mathrm{SOC}) \cdot \mathbb{I}_{T<T_{\mathrm{threshold}}} \tag{26}$$

We encode these as 15 Bayesian priors in the VAE training.

### 5.2 Training Data Generation

Training data generated from open-source physics simulators:

- **PyBaMM** [11]: Doyle-Fuller-Newman electrochemical model
- **OpenFOAM Electrochemistry**: 3D thermal-electrochemical coupling
- **liionpack**: Multi-cell pack simulations

**Dataset**: 50,000+ synthetic cycles spanning:

- Multiple chemistries (LFP, NMC)
- Temperature range: 15-45°C
- C-rates: 0.5C – 2.0C
- Complete ground truth for all internal states

### 5.3 Architecture Details

**Encoder**: 16 → 128 → 64 → 32 (latent)
**Decoder**: 32 → 64 → 128 → 16
**Physics weights** $\Phi_\alpha$ (normalized):

- SOH: 3.0 (target variable)
- Internal resistance: 2.5 (primary health indicator)
- Cell voltages: 2.0 (safety critical)
- Capacity: 1.5 (health measure)
- Temperature: 1.0 (thermal stress)

### 5.4 Results

**Inference Speed**: 20-200× faster than online physics simulation

**Accuracy**:

- SOH MAE: $0.008 \pm 0.003$
- Resistance MAE: $3\,\mathrm{m}\Omega \pm 1\,\mathrm{m}\Omega$
- $\Delta$SOH/cycle MAE: 0.0001%/cycle

**Physics Consistency**: 100% of predictions satisfy:

- Monotonic SOH decline
- Energy conservation
- Physical bounds (voltage, temperature)

### 5.5 Component Diagnostic Example

Consider two batteries both at 85% SOH:

**Battery A (Normal)**:

- M-W Distance: 1.2 [NORMAL]
- Component breakdown: Capacity 60%, Thermal 20%, Electrical 15%
- Diagnosis: Uniform calendar aging

**Battery B (Anomalous)**:

- M-W Distance: 2.3 [ELEVATED]
- Component breakdown: Electrical 78%, Thermal 15%, Capacity 4%
- Diagnosis: Anomalous resistance growth → Inspect within 30 days

Traditional metrics would treat these identically. Our framework reveals fundamentally different degradation patterns.

## 6 Validation: OT Cybersecurity

### 6.1 Domain Physics

Operational technology (OT) security exhibits analogous mathematical structure:

1. **Exposure Time "Activation Energy"**:

$$f_{\text{exp}}(t) = \exp\left(E_{\text{exp}} \cdot \left(\frac{t}{t_{\text{ref}}} - 1\right)\right) \tag{27}$$

2. **Vulnerability Accumulation (Parabolic Law)**:

$$V(t) = k_{\text{vuln}}\sqrt{t} + \text{CVE}_{\text{critical}} \cdot w_{\text{severity}} \tag{28}$$

3. **Network Security Stress**:

$$\sigma_{\text{net}} = (1 - k_{\text{seg}} \cdot S) \cdot (1 + k_{\text{remote}} \cdot R) \cdot (1 + w_{\text{proto}} \cdot P) \tag{29}$$

4. **Attack Propagation**:

$$P_{\text{lateral}} = k_{\text{lateral}} \cdot \log(1 + \text{assets}) \cdot (1 - \text{segmentation}) \tag{30}$$

**Remarkable observation**: Despite completely different physics (electrochemistry vs cybersecurity), we see:

- Same $\sqrt{t}$ growth law (SEI vs vulnerabilities)
- Same exponential acceleration (Arrhenius vs exposure)
- Same stress multiplication (mechanical vs network)

This suggests universal mathematical structure underlying constrained dynamics.

## 6.2 Training Data

Synthetic OT environments with:

- 22 state dimensions
- Industry sectors: Energy, Manufacturing, Transportation
- Attack scenarios: Ransomware, lateral movement, process manipulation
- 10,000+ organization-year trajectories

## 6.3 Results

**Security Health Prediction**:

- Health score MAE: $0.012 \pm 0.004$
- Attack likelihood AUC: 0.89
- False positive rate: 5%

**Component Breakdown**: Organization with 60% security health:

- M-W Distance: 2.8 [HIGH]
- Vulnerabilities: 65% (unpatched CVEs)
- Network: 20% (poor segmentation)
- Access: 10% (weak authentication)
- Recommended: Emergency patching within 14 days

**Attack Likelihood Predictions**:

- Ransomware: $28\% \pm 4\%$
- Lateral movement: $31\% \pm 5\%$
- Process manipulation: $15\% \pm 3\%$

Table 2: Mathematical convergence across domains

| Property | Battery | Cyber |
|---|---|---|
| Growth law | $\sqrt{t}$ SEI | $\sqrt{t}$ CVE |
| Acceleration | Arrhenius | Exposure |
| Stress factor | Mechanical | Network |
| Manifold dim | 32-D | 32-D |
| M-W properties | ✓ | ✓ |

# 7 Discussion

## 7.1 A Unified Geometric Framework

The convergence of independently-derived geometric structures across batteries and cybersecurity provides empirical evidence for our central thesis: physics constraints naturally induce Riemannian geometry, independent of specific domain.

This is not coincidence—it reflects a deeper principle that our framework captures.

## 7.2 Generalizing the Riemannian Sector: GR as One Application

Our central theoretical contribution is generalizing the Riemannian sector of General Relativity's mathematics to arbitrary physics-constrained inference.

**GR's Riemannian Sector** (What We Generalize):

- Positive-definite metric tensors $g_{ij}$ on spatial hypersurfaces
- Synge's world function $\Omega(P, Q)$ for Riemannian manifolds
- Geodesic distance computations
- Christoffel symbols $\Gamma^k_{ij}$ and curvature tensors
- Van Vleck determinant $\Delta(P, Q)$

**GR's Lorentzian Structure** (Beyond Current Scope):

- Indefinite signature $(-, +, +, +)$ for spacetime
- Timelike vs spacelike separation
- Causal structure and light cones
- Time ordering and chronological future/past
- Lorentzian distance (can be imaginary)

**Our Generalization**:

- $N$-dimensional Riemannian state manifolds (not just 3D spatial slices)
- Metrics $g_{ij}$ learned via Bayesian inference (not just from Einstein equations)
- Applied to arbitrary physics constraints (not just gravity)
- Automatic differentiation for computation (not manual tensor calculus)
- Real-time inference (not offline PDE solving)

**The Universal Structure (Present in Both)**:
The Riemannian geometric formalism admits world functions $\Omega(\mathbf{x}, \mathbf{x}')$ satisfying:

1. Coincidence limit: $\Omega(\mathbf{x}, \mathbf{x}') \to 0$ as $\mathbf{x}' \to \mathbf{x}$
2. Geodesic correspondence: $\nabla\Omega$ yields geodesic tangents
3. Parameterization invariance: Independent of curve parameterization
4. Van Vleck determinant: $\Delta(\mathbf{x}, \mathbf{x}')$ measures geodesic focusing

These properties hold whether the manifold represents spatial slices of GR spacetime, battery degradation states, or cybersecurity attack surfaces.

**Precise Statement**:

**What We Generalize**: Riemannian geometry (positive-definite metrics, Synge world function, geodesic geometry, curvature tensors) to arbitrary $N$-dimensional physics-constrained manifolds

**Outside Present Scope**: Lorentzian structure (causality, time ordering, indefinite signature)

**Evidence**: Identical Riemannian structures ($\sqrt{t}$ growth, exponential acceleration, stress accumulation) appear in electrochemistry, cybersecurity, and spatial GR

**Theoretical Significance**:
Riemannian geometry is not specific to spatial slices of curved spacetime—it's a universal framework for constrained inference. GR applies it to gravity; we apply it to batteries, cybersecurity, and any physics-constrained system.

### 7.3 Scope and Boundaries of the Generalization

To ensure clarity, we explicitly state the scope and boundaries of our generalization:

**What We Generalize**:

1. **Riemannian Geometry**: Positive-definite metric tensors, geodesic distances, Synge world function on Riemannian manifolds, Christoffel symbols, curvature tensors
2. **Mathematical Formalism**: The computational and inferential framework for working with curved manifolds—variational principles, parallel transport, covariant derivatives
3. **Applicability**: From spatial geometry in GR to arbitrary physics-constrained state manifolds (batteries, cybersecurity, etc.)

**Outside Present Scope**:

1. **Lorentzian Signature**: Indefinite metric $(-, +, +, +)$ with timelike/spacelike distinction
2. **Causal Structure**: Light cones, chronological ordering (addressed in Section 7.5)
3. **Einstein Dynamics**: Field equations $G_{\mu\nu} = 8\pi T_{\mu\nu}$—we learn metrics from data
4. **Diffeomorphism Invariance**: Our learned metrics are prior-dependent

**Applications Not Requiring Causality**:
Most constrained inference problems have natural time evolution without needing Lorentzian structure:

- Battery degradation: Monotonic SOH decline

- Cybersecurity: Accumulating vulnerabilities
- Process monitoring: Forward-only state transitions

For these, Riemannian geometry is sufficient and computationally simpler.

### 7.4 Computational Advantages

**Real-time capability**:

- GR: Numerical relativity requires supercomputers, weeks of calculation
- M-W: Inference in milliseconds on standard hardware

**Scalability**:

- GR: 20 known exact solutions (high symmetry required)
- M-W: Works for any learnable physics constraints

**Deployment**:

- GR: Theoretical physics, LIGO data analysis
- M-W: Battery management systems, cybersecurity platforms, nuclear monitoring

### 7.5 Limitations and Future Work

**Current limitations**:

1. **Riemannian Restriction**: Our current framework employs Riemannian (positive-definite) metrics:
$$g_{ij}(\mathbf{z}) = \sum_{\alpha} \frac{\partial D^\alpha}{\partial z^i} \frac{\partial D^\alpha}{\partial z^j} \Phi_\alpha > 0 \tag{31}$$
This differs from General Relativity's Lorentzian signature: $g_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$

   **Implications**:

   - No causal structure: Our manifolds lack the timelike/spacelike distinction central to relativistic physics
   - No invariant time direction: Geodesics don't distinguish "forward" vs "backward" propagation
   - Restricted GR analogy: We can only claim analogy to the *Riemannian sector* of GR (spatial hypersurfaces), not full spacetime

   **Why This Doesn't Invalidate Our Claims**:

   - Most state estimation problems don't require causality: Battery degradation and cybersecurity have natural time evolution without needing Lorentzian structure
   - Synge's properties still hold: Theorems 1-4 apply to Riemannian manifolds; Lorentzian extension would preserve these properties
   - Our universality claim survives: The geometric inference structure exists in both Riemannian and pseudo-Riemannian settings

2. VAE only learns manifold within training distribution
3. Full Riemannian geodesics computationally expensive (use linear approximation)

4. Physics weights $\Phi_\alpha$ require domain expertise

5. Regulatory certification needed for safety-critical deployment

**Future directions**:

1. **Pseudo-Riemannian Extension**:
   - Modified metric construction with $\eta_\alpha \in \{-1, +1\}$ allowing indefinite signatures
   - Time-series applications with backward/forward distinction
   - Causal inference in dynamical systems
   - Full GR analogy with Lorentzian structure

2. **Exact geodesic computation**: GPU-accelerated numerical integration

3. **End-to-end metric learning**: Learn $g_{ij}(\mathbf{z})$ directly

4. **Formal verification**: Prove physics constraints hold with probability $1 - \delta$

5. **Connection to quantum**: Explore relationship to path integrals via Van Vleck determinant

6. **Information geometry**: Relate to Fisher information metrics

7. **More domains**: Nuclear, aerospace, chemical, medical

### 7.6 Broader Impact

**Scientific**: Opens new research direction at intersection of:

- Differential geometry
- Bayesian deep learning
- Domain-specific physics
- Information theory

**Engineering**: Practical framework for safety-critical monitoring:

- Battery management ($50B+ market)
- OT cybersecurity ($20B+ market)
- Nuclear safety (regulatory compliance)
- Aerospace health monitoring (mission-critical)

**Philosophical**: Suggests geometric principles may be more universal than previously recognized, extending beyond spacetime physics to constrained inference generally.

## 8 Conclusion

We have presented the Modak-Walawalkar framework—a generalization of the Riemannian sector of General Relativity's mathematics to arbitrary physics-constrained state estimation. Our key contributions are:

1. **Theoretical**: Proof that learned Riemannian manifolds from Bayesian VAEs satisfy Synge-type world function properties (coincidence limits, geodesic correspondence, parameterization invariance)—Theorem 4 establishes that this Riemannian geometric structure generalizes beyond spatial GR

2. **Computational**: Replacement of tensor calculus with automatic differentiation, making Riemannian geometric methods tractable for real-world deployment

3. **Universal**: Demonstration across electrochemistry and cybersecurity that identical Riemannian structures emerge from entirely different physics, validating the generalization

4. **Practical**: Deployment-ready system achieving 20-200$\times$ speedup with formal uncertainty guarantees

The convergence with Synge's Riemannian formulation—arrived at independently from first principles of physics-constrained estimation—provides strong evidence for our central thesis.

We generalize the Riemannian sector of GR's mathematics: positive-definite metrics, geodesic distances, world functions, and curvature computations apply to any constrained manifold—spatial slices of GR spacetime, battery state spaces, cybersecurity attack surfaces. The Lorentzian extension (causality, time ordering) is acknowledged as a separate mathematical layer requiring distinct treatment, addressed in future work.

General Relativity's spatial geometry represents one application of Riemannian inference: 3D hypersurfaces with metrics constrained by Einstein dynamics. Our framework reveals the broader pattern: $N$-dimensional Riemannian manifolds with learned metrics and arbitrary physics priors. The mathematical convergence—identical world function properties across spatial GR, electrochemistry, and cybersecurity—demonstrates that Riemannian geometric inference is universal, not domain-specific.

This work establishes that the Riemannian mathematical framework developed for spatial relativity generalizes to constrained inference across diverse domains, opening new directions at the intersection of differential geometry, Bayesian learning, and physics-informed systems with transformative potential for safety-critical engineering.

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